

# Vihaan Gupta

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## SUMMARY

Third-year B.Tech Computer Science student and robotics software engineer working across computer vision, machine learning and autonomous robotics. Strong mathematical grounding in machine learning and deep learning, applied to perception, manipulation and navigation on real robots, from ROS 2 and Isaac Sim to NVIDIA Jetson hardware. Seeking internships in robotics, computer vision and AI/ML.

## COMPETITIONS & ACHIEVEMENTS

**Best Innovation Award, Caterpillar Autonomy Challenge** IIT Madras

- Autonomous excavation system built with The MaRS Club: designed control strategies and trajectory planning for a 5-DOF manipulator (excavation and berm construction).
- Integrated SLAM and navigation pipelines and interfaced an Intel RealSense camera for real-time perception.

**Rank 10, International Rover Challenge & Rank 9, International Space Drone Challenge** SPROS, 2026

- Worked on the rover's robotic manipulator, including Cartesian control, and on YOLO-based object detection for task execution.
- Worked with RealSense cameras, IMUs, encoders and Jetson AGX Orin; contributed to perception, navigation and system integration.

## SKILLS

**Robotics:** ROS 2, MoveIt2, Gazebo, NVIDIA Isaac Sim, Nav2, SLAM (RTAB-Map), manipulator control and trajectory planning

**Robot Mathematics:** rigid-body kinematics, Jacobians, twists, linear algebra

**Computer Vision:** OpenCV, Open3D (point clouds), depth cameras, YOLO object detection

**ML / Deep Learning:** PyTorch, ML and DL architectures, NumPy, Pandas, Matplotlib

**Programming:** C, C++, Python; OOP and data structures, MySQL, HTML/CSS

**Systems & Hardware:** Linux, operating systems, Docker; Jetson AGX Orin and Orin Nano, Raspberry Pi, Arduino, ZED 2i, RealSense, IMUs and encoders; basic 3D printing

**Robots Worked On:** custom 5-DOF manipulator, Mars rover, differential-drive autonomous robot, SO-101, line followers and RC robots

## EXPERIENCE & PROJECTS

**Member, The MaRS Club (Robotics Society), IIITDM Kancheepuram** 2024 – Present

- Work on manipulation, perception and navigation for the club's rover, manipulator and competition robots.

**Autonomous Carrom-Playing Robot, Research Internship** Under Dr. Nagamanikandan G.

- Robot that plays carrom physically against humans: YOLO-based perception tracks coins and striker, and a strategy algorithm selects shots.

**Autonomous Manipulator, UR10e in NVIDIA Isaac Sim** In progress | ROS 2, Isaac Sim

- Developing a pick-and-place pipeline that brings perception, planning and control into one loop, with hardware deployment as the next step.

**Minibot, Autonomous Navigation** Nav2, RTAB-Map, Jetson Orin Nano

- Differential-drive indoor robot navigating autonomously with Nav2 and RTAB-Map SLAM using a ZED 2i camera.

**Manipulation, ROS 2 + Gazebo + MoveIt2** ROS 2, MoveIt2

- Cartesian and joint-space control, collision-aware planning and trajectory execution for a manipulator.

**Perception and Deep Learning** PyTorch, DINO, Open3D

- Object pose estimation for robotic manipulation, and DINO-based perception packaged for ROS 2.
- PyTorch models for surgical image segmentation and steel surface defect classification.

## EDUCATION

**B.Tech in Computer Science and Engineering, IIITDM Kancheepuram** 2024 – Present

**Class XII (PCM), St. Joseph's Co-ed School, Bhopal** 2024

**Certification:** Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford Online (Coursera), 2025

## LANGUAGES

English, Hindi